



MAKU DIGITAL COOPERATIVE PLATFORM

Software Development & Digital Transformation Proposal

Prepared for:
Merti Animal Key Users (MAKU)
Merti, Isiolo County, Kenya

Prepared by:
Trends CORE Solutions
We build solutions for business.

Digital platform proposal

Contents

A concise guide to the proposal and its major sections.

01	Executive Summary	4
02	Our Understanding of MAKU	6
03	Proposed Digital Platform	7
04	Management & Operations	8
05	Finance, Documents & Partnerships	16
06	Public Website & Digital Identity	39
07	Technology & Architecture	49
08	Implementation, Training & Support	51
09	Investment & Running Costs	62
10	Governance, Success & Conclusion	72

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1. Executive Summary

Merti Animal Key Users (MAKU) is a farmer-led cooperative society based in Merti, Isiolo County, Northern Kenya.

MAKU brings together farmers through smaller Common Interest Groups (CIGs) and is developing a growing range of agricultural and livestock-related activities, including:

- Livestock marketing
- Beekeeping and honey production
- Hive production and sales
- Skins and hides
- Poultry farming
- Bones and horns
- Environmental conservation and seedling sales
- Feedlot management
- Dairy milk aggregation and value addition
- Farmer/member services

As these activities grow, MAKU's administrative, financial, documentation and reporting requirements will also grow.

The proposed MAKU Digital Cooperative Platform will bring these activities into one simple digital system.

Instead of having information scattered across:

exercise books → manual vouchers → WhatsApp → Excel → files → receipts → physical documents

MAKU will have a centralized system where authorized staff can:

Register → Record → Approve → Pay → Sell → Report → Communicate → Archive

The system will also provide MAKU with a professional public website that makes the organization easier to discover by:

- Farmers
- Members
- NGOs
- Government agencies
- Development partners
- Donors
- Researchers
- Buyers
- Investors

- Other potential partners

The objective is not simply to computerize MAKU's current work.

The objective is to create a foundation on which MAKU can grow.

2. Our Understanding of MAKU

We understand MAKU as more than a livestock market.

It is evolving into a farmer aggregation and value-addition organization serving communities around Merti.

The digital platform should therefore reflect the different activities of the cooperative.

Current / Active

Activity	Digital Support
Livestock Marketing	✓
Beekeeping	✓
Hive Manufacturing & Sales	✓
Environmental Conservation	✓
Seedling Sales	✓
Feedlot	✓
Member/CIG Management	✓
Finance & Office Management	✓
Documentation	✓
Partnerships & Grants	✓

Developing / Upcoming

Activity	Digital Support
Hides & Skins	✓
Poultry	✓
Bones & Horns	✓
Dairy/Milk Aggregation	✓
Honey Production	✓
Value Addition	✓

The platform will therefore be modular, allowing MAKU to activate additional business areas as they become operational.

3. Proposed Solution

MAKU Digital Cooperative Platform

We propose developing a centralized web-based system consisting of two connected parts:

A. MAKU Management System

The private system used by MAKU staff, officials and authorized users.

B. MAKU Public Website

The public-facing website explaining who MAKU is, what it does, where it is located and how development partners and other stakeholders can engage with the organization.

Together:

MAKU Website + MAKU Management System = MAKU Digital Platform

4. MAKU Management System

4.1 Member & Farmer Register

The member register will become one of the most important assets of MAKU.

Each member will have a digital profile.

Member information can include:

- Full name
- Member number
- National ID/reference
- Phone number
- Location
- Village
- CIG
- Gender
- Next of kin
- Membership date
- Membership status
- Activities
- Livestock owned
- Shares
- Membership contribution
- Payment history
- Transactions with MAKU
- Documents
- Notes

Membership structure

The system will accommodate the current contribution model:

Membership contribution: KES 1,000

Share contribution: KES 5,000

The system can automatically show:

Member → Amount paid → Balance → Shares → Transaction history

5. CIG Management

MAKU currently has approximately 16 Common Interest Groups (CIGs).

The system will allow MAKU to create and manage these groups.

Each CIG can have:

- CIG name
- Chairperson
- Secretary
- Treasurer
- Members
- Location
- Activities
- Meetings
- Contributions
- Livestock aggregation
- Sales
- Group documents
- Reports

Example

Merti Camel & Livestock CIG

Members: 24

Livestock aggregated: 87

Animals sold: 52

Total sales: KES XXX

MAKU charges: KES XXX

Member proceeds: KES XXX

This gives MAKU a much clearer picture of what is happening at grassroots level.

6. Livestock Marketing Module

This should be the core module of the first phase.

The system will manage the complete livestock journey.

From:

Farmer → Animal → Aggregation → Sorting → Feedlot → Sale → Buyer → Payment → Farmer

The system should record:

Animal intake

- Owner/member
- CIG
- Species
- Breed/type
- Sex
- Age
- Identification/tag
- Condition
- Estimated weight
- Purchase/aggregation value
- Date received
- Location
- Photos where necessary

Aggregation

The system can identify:

- Number of animals collected
- Farmers supplying animals
- CIG supplying animals
- Animals currently at the feedlot
- Animals sold
- Animals remaining

Sale

Record:

- Buyer
- Date
- Animal
- Seller/member
- Sale price
- MAKU service charge
- Other expenses
- Net amount payable to farmer
- Payment method
- Payment status

7. MAKU Livestock Service Charge

The current arrangement is:

KES 200 per animal sold on behalf of a farmer.

The system will automatically calculate this.

For example:

*Animal sold: KES 45,000
MAKU service charge: KES 200
Other approved expenses: KES 500
Farmer proceeds: KES 44,300*

This reduces calculation errors and creates an auditable trail.

8. Feedlot Management

The feedlot should have its own digital section.

MAKU can monitor:

- Animals currently in feedlot
- Entry date
- Expected exit date
- Feed consumed
- Watering
- Health observations
- Weight changes
- Expenses
- Feed purchases
- Water expenses
- Labour
- Mortality/losses
- Sale status

Feedlot dashboard

Management could see:

Animals Present: 86 Animals Sold: 32 Average Days: 18 Feed Cost: KES XXX Water Cost: KES XXX
Other Costs: KES XXX

This will become particularly valuable once the borehole project is operational.

9. Borehole & Water Voucher Management

The current manual process is an excellent candidate for digitization.

Currently:

Animals watered → borehole owner issues voucher → voucher goes to MAKU → MAKU pays

The system can introduce a Water Voucher Register.

Each voucher can have:

- Voucher number
- Date
- Borehole
- Borehole owner
- Number of animals
- Rate
- Amount
- Feedlot/CIG
- Authorized person
- Payment status
- Supporting document/photo

Eventually, authorized users could generate a digital voucher directly from the system.

This creates a proper audit trail.

10. Livestock Purchases

MAKU will need a dedicated purchasing process.

The system should distinguish between:

Purchase

MAKU purchases livestock directly.

and

Marketing on behalf

MAKU sells livestock belonging to members/farmers.

This distinction is important for accounting and reporting.

Purchases can capture:

- Supplier
- Farmer
- Animal
- Quantity
- Price
- Negotiated price
- Transport
- Other costs
- Payment
- Approval
- Receipt
- Supporting documents

11. Sales Management

The sales module will provide daily, weekly and monthly reporting.

Example reports

Daily Sales

- Animals sold
- Buyers
- Total sales
- MAKU charges
- Farmer proceeds

Weekly Sales

- Number of animals
- Total sales
- Total service charges
- Total expenses
- Net proceeds

Monthly Sales

- Total animals
- Total turnover
- Average selling price
- CIG performance
- Member participation
- Buyer analysis

12. Payments & Finance

MAKU needs a simple but controlled internal financial system.

It should manage:

Cash

- Petty cash
- Cash receipts
- Cash payments
- Cashbook
- Expenses

Bank

- Bank transactions
- Deposits
- Withdrawals
- Transfers
- Reconciliation

M-Pesa

- Member payments
 - Farmer payments
 - Supplier payments
 - Sales proceeds
 - Other transactions
-

13. Petty Cash

The system will provide a petty cash register.

Every transaction should capture:

Request → Approval → Payment → Receipt → Recording

For example:

Petty Cash Request

Purpose: Stationery Amount: KES 1,500 Requested by: Secretary Approved by: Authorized Officer

Paid by: Cashier Receipt: Attached

This protects MAKU from undocumented expenditure.

14. Expenses & Procurement

A proper procurement workflow can be included.

Example

Staff member requests:

10 bags of livestock feed

System records:

Request → Approval → Supplier → Purchase → Delivery → Payment → Document

MAKU will therefore have a history of:

*What was bought?
Who requested it?
Who approved it?
Who supplied it?
How much did it cost?
Was it delivered?
Was it paid?*

15. Salaries & Staff Management

A basic staff management module can manage:

- Staff register
- Position
- Employment documents
- Salary
- Allowances
- Advances
- Deductions
- Payment records
- Leave
- Attendance

The system can later integrate with a more advanced payroll system if MAKU requires it.

16. Procurement & Supplier Register

MAKU will eventually deal with many suppliers.

The system will maintain a supplier database.

Examples:

- Feed suppliers
- Borehole operators
- Transporters
- Veterinary suppliers
- Stationery suppliers
- Construction suppliers
- Hive material suppliers
- Seedling suppliers
- Equipment suppliers

Each supplier can have:

- Contact details
- KRA/PIN where applicable
- Bank/M-Pesa details
- Products/services
- Contracts
- Documents
- Transaction history

17. Honey & Beekeeping Module

MAKU is already making hives and developing honey farming.

The system should be ready for this.

Hive production

Track:

- Hive type
- Quantity produced
- Cost
- Materials
- Labour
- Selling price
- Buyer
- Stock

Honey production

Track:

- Apiary
- Hive
- Location
- Colony status
- Inspection
- Harvest
- Quantity
- Quality
- Sales

This can later develop into a complete Honey Enterprise Management System.

18. Environmental Conservation

This is an interesting area where MAKU can create measurable impact.

Members planting seedlings can be digitally recorded.

The system can track:

- Member
- CIG
- Seedlings received
- Seedlings planted
- Species
- Location
- Date
- Survival rate
- Seedlings sold
- Revenue

Impact dashboard

For example:

*12,450 seedlings planted
312 farmers involved
16 CIGs
participating
8,900 trees surviving*

This type of information can become extremely valuable when approaching development partners.

19. Hides & Skins

When this activity becomes operational, the module can manage:

- Collection
 - Farmer
 - Animal origin
 - Type
 - Quantity
 - Quality/grade
 - Storage
 - Buyer
 - Sales
 - Payments
-

20. Poultry

The future poultry module can manage:

- Farmers
 - Birds
 - Breeds
 - Production
 - Eggs
 - Feed
 - Mortality
 - Sales
 - Buyers
 - Revenue
-

21. Bones & Horns

A future module can manage:

- Collection
 - Source
 - Quantity
 - Processing
 - Storage
 - Buyers
 - Sales
 - Revenue
-

22. Dairy & Milk Aggregation

The proposed dairy business could eventually operate through:

Farmer → Milk Collection Point → MAKU → Processing/Value Addition → Buyer

The system can record:

- Farmer
- Collection point
- Morning/evening quantity
- Quality
- Price
- Farmer payment
- Total milk
- Spoilage
- Buyer
- Sales

This can eventually become one of MAKU's major aggregation systems.

23. Complete Document Centre

This is one of the areas I recommend MAKU takes very seriously.

Instead of keeping important documents in physical files, WhatsApp and individual computers, MAKU should have a Digital Document Centre.

Documents can include:

Corporate

- Registration certificates
- Constitution
- Minutes
- AGM documents
- Board resolutions

Financial

- Audit reports
- Bank documents
- Financial statements
- Receipts
- Payment vouchers

Compliance

- KRA documents
- Tax compliance
- Licences
- Certifications
- Regulatory documents

Projects

- Feedlot documents
- Borehole documents
- Aggregation documents
- Environmental projects

Partnerships

- MOUs
- Grant agreements
- NGO correspondence
- Project proposals
- Contracts

24. Smart Document Search

Instead of asking:

"Who has the 2025 audit report?"

A MAKU user can search:

Audit Report 2025

and retrieve it.

Documents can have:

- Category
- Year
- Project
- Organisation
- Expiry date
- Owner
- Confidentiality
- Tags

The system can also alert authorized users:

KRA Tax Compliance Certificate expires in 30 days.

25. Document Creation & Communication Centre

This is where we can take the system further.

MAKU should be able to create documents from templates.

For example:

Select:

Letter → NGO Partnership Request

The system automatically inserts:

- MAKU name
- Address
- Contact details
- Date
- Recipient
- Subject
- Authorized signatory

The document can then be:

Create → Review → Approve → PDF → Send

This reduces repetitive office work.

26. Communication Module

MAKU will need a central communication system.

Communication channels can include:

- Email
- SMS
- WhatsApp
- Website forms
- Internal notices

Examples

Send members:

"Livestock collection will take place on Friday."

Send CIG leaders:

"Your group's animals have been scheduled for aggregation."

Send NGO:

"MAKU has submitted the attached feedlot proposal."

The system can maintain a record of communications.

27. NGO & Development Partner Management

This is an area where I recommend MAKU goes beyond a normal cooperative system.

Create a Partners & Opportunities Register.

For every NGO/donor/partner:

- Organization
- Contact person
- Email
- Phone
- Website
- Area of interest
- Funding opportunities
- Previous engagement
- Proposals submitted
- Status
- Documents
- Follow-up dates

Potential partners can include organizations such as:

- Mercy Corps
- FSRP
- WFP
- World Food Programme
- Government programmes
- NGOs
- Development agencies
- Foundations
- Research institutions
- Private sector partners

The system should not assume that an organization is currently funding MAKU. Instead, it should distinguish:

Prospect → Contacted → Proposal Submitted → Under Review → Approved → Active → Completed

28. Grants & Proposal Pipeline

This could become one of MAKU's most valuable administrative tools.

For every opportunity:

Opportunity

↓ Requirements

↓ Proposal

↓ Budget

↓ Submission

↓ Follow-up

↓ Decision

↓ Project

↓ Reporting

This prevents opportunities from getting lost in WhatsApp or someone's notebook.

29. MAKU Project Management

Every major initiative can become a project.

Examples:

Merti Feedlot Expansion

Budget: KES XXX Funding: XXX Status: In Progress

Borehole Project

Status: Construction

Honey Production Project

Status: Development

Tree Seedling Programme

Status: Active

Each project can have:

- Budget
- Activities
- Timeline
- Responsible persons
- Documents
- Photos
- Funding
- Reports
- Milestones

30. Roles & Permissions

Because MAKU has several officials, users should not all have access to everything.

Suggested roles:

System Administrator

Full system management.

Chairman

Strategic oversight and reports.

Secretary General

Administration, correspondence and documentation.

Treasurer

Finance and financial reports.

Cashier

Payments and receipts.

Procurement Officer

Purchasing and suppliers.

Livestock Officer

Animals, aggregation and sales.

CIG Officer

CIG and member management.

Project Officer

Projects, NGOs and grants.

Records Officer

Document management.

Auditor / Read-only Auditor

Access to reports and audit trails.

The system can later support additional roles.

31. Approval System

An important principle should be:

The person who requests should not necessarily be the person who approves and pays.

For example:

Purchase Request

Staff → Procurement → Approval → Purchase → Delivery → Payment

This gives MAKU stronger internal controls.

32. Audit Trail

Every important action should leave a record.

The system should be able to answer:

<i>Who created this transaction?</i>
<i>Who changed it?</i>
<i>When was it changed?</i>
<i>Who approved it?</i>
<i>Who paid it?</i>
<i>Which document supports it?</i>

This is particularly important for cooperative governance, audits and donor-funded projects.

33. Management Dashboard

When the Chairman or Secretary logs in, they should immediately understand the state of MAKU.

Example:

MAKU TODAY

Members 1,248

CIGs 16

Animals in Feedlot 86

Animals Sold This Month 143

Livestock Sales KES 6.8M

MAKU Service Charges KES 28,600

Pending Payments KES XXX

Active Projects 7

Pending Approvals 12

Documents Expiring Soon 3

Grant Opportunities 5

This turns data into management information.

34. Reports

Reports should be available by:

Daily

- Cash
- Sales
- Purchases
- Expenses
- Animals
- Payments

Weekly

- Livestock sales
- Member activity
- CIG activity
- Feedlot
- Cashflow

Monthly

- Income
- Expenses
- Sales
- Purchases
- Membership
- CIG performance
- Projects
- Partner activities

Annual

- Annual membership
- Shares
- Livestock turnover
- Projects
- Financial reports
- Environmental impact
- Enterprise performance

Reports should be downloadable as:

PDF / Excel / CSV

35. Public MAKU Website

The website should not merely say:

"Welcome to MAKU."

It should work as a digital front door to the cooperative.

Proposed Website Sections

Home

A strong statement about MAKU.

Example direction:

Empowering Farmers. Connecting Markets. Building Resilient Communities.

About MAKU

- Who we are
- Our history
- Our vision
- Mission
- Leadership
- Membership

What We Do

Interactive enterprise sections:

☐ Livestock Marketing ☐ Beekeeping ☐ Environmental Conservation ☐ Feedlot Management ☐ Dairy ☐ Poultry ☐ Hides & Skins ☐ Agricultural Value Addition

36. MAKU Impact Page

This is where the website becomes more powerful for partners.

Display measurable information such as:

1,200+ Farmers

16 CIGs

47,000+ Animals Aggregated

12,000+ Seedlings Planted

KES XX Million Farmer Transactions

Figures would be populated from the management system once reliable data exists.

37. NGO & Partner Page

A dedicated page:

Partner With MAKU

Explain:

- What MAKU does
- Where we operate
- Farmer reach
- CIG structure
- Current projects
- Development needs
- Partnership opportunities
- Contact details

Then:

Current Opportunities

- Feedlot expansion
- Borehole development
- Honey production
- Dairy aggregation
- Environmental conservation
- Poultry development
- Value addition

This makes MAKU easier for development organizations to understand.

38. MAKU Location & GIS

This is particularly important given MAKU's location.

The website should clearly show:

MAKU Headquarters

Merti, Isiolo County, Kenya

with an interactive map.

We can include:

- GPS coordinates
- Google Maps location
- What3Words/location reference if useful
- Directions
- Nearby towns
- Contact information

The coordinates should be supplied/verified by MAKU rather than guessed.

39. "Find MAKU" Feature

I would go a step further.

Create a dedicated:

FIND MAKU

page.

It can show:

☐ MAKU office ☐ Feedlot ☐ Borehole ☐ CIG locations ☐ Collection points ☐ Apiaries ☐ Seedling sites

As the organization grows, MAKU can build its own small operational map.

That could become very useful to development partners and researchers.

40. Partner Download Centre

The website can have a controlled section where potential partners can download:

- MAKU profile
- Organization registration
- Annual reports
- Project briefs
- Concept notes
- Audited reports
- Contact information
- Capability statement

Some documents can remain public while sensitive documents remain private.

41. Website Enquiries

Visitors can submit:

I want to:

- Partner with MAKU
- Buy livestock
- Supply products
- Support a project
- Become a member
- Request information
- Contact management

The enquiry automatically enters the MAKU system.

This means:

Website → CRM/Communication → Follow-up → Record

rather than simply receiving an email that may be forgotten.

42. Mobile Friendly

The system should be designed mobile-first.

This is important because MAKU's activities happen outside an office.

A livestock officer may be:

- At a market
- At a farm
- At the feedlot
- At a CIG
- At a collection point

They should be able to use a phone.

43. Offline-Friendly Operation

Because MAKU operates in Northern Kenya, connectivity should be considered from the beginning.

Where practical, field forms should be designed to tolerate poor connectivity.

For example:

Record animal → Save locally → Synchronize when internet returns.

This should be included particularly for:

- Livestock intake
- Member registration
- CIG activities
- Feedlot records
- Environmental projects
- Field data

44. Security

The system will implement:

- User accounts
- Password security
- Role-based permissions
- Approval controls
- Audit logs
- Database backups
- Document access controls
- Secure connections
- Regular backups

Sensitive financial and member information should never be publicly accessible.

45. Proposed Technology Stack

We recommend a modern web technology stack.

Frontend

React + TypeScript

Why?

- Fast
- Modern
- Responsive
- Scalable
- Excellent for dashboards
- Can support future mobile/PWA capabilities

Backend

Node.js + TypeScript

Why?

- Reliable modern backend
- Same language across frontend/backend
- Efficient development
- Strong ecosystem
- Easy to maintain as a sole developer

Database

PostgreSQL

Why?

MAKU will have highly connected information:

Member → CIG → Animal → Sale → Payment → Document → Project

PostgreSQL is well suited for this relational structure.

Document Storage

Secure cloud/object storage for documents and attachments.

Maps

Google Maps / OpenStreetMap-based mapping depending on requirements and cost.

Notifications

Depending on implementation:

- Email
- SMS
- WhatsApp
- M-Pesa integrations

46. Why This Technology?

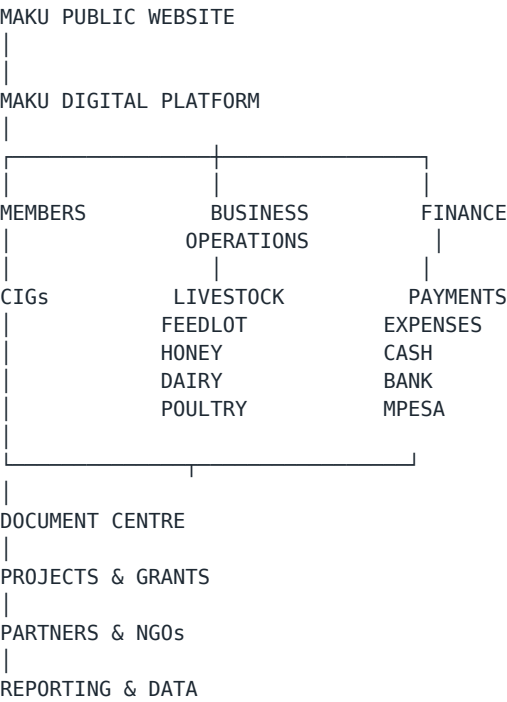
The platform should not be built around a technology trend.

It should be built around:

Maintainability + Security + Scalability + Cost + Availability of developers

As MAKU grows, the platform should remain maintainable without being locked into an unnecessarily expensive proprietary ecosystem.

47. Proposed Architecture



48. Implementation Approach

Because MAKU has many activities, we recommend phased implementation rather than trying to build everything at once.

Phase 1 — Foundation

4-6 weeks

- User management
 - Roles & permissions
 - Member register
 - CIG management
 - Membership contributions
 - Share contributions
 - Basic finance
 - Document Centre
 - Public website
 - MAKU profile
 - Contact management
 - Basic reporting
-

49. Phase 2 — Livestock Operations

4-6 weeks

- Livestock intake
 - Farmer aggregation
 - Animal records
 - Livestock purchases
 - Livestock sales
 - KES 200 service charge
 - Farmer settlement
 - Buyer management
 - Feedlot
 - Water vouchers
 - Feed management
 - Livestock reports
-

50. Phase 3 — Finance & Administration

3-5 weeks

- Procurement
 - Suppliers
 - Petty cash
 - Expenses
 - Payments
 - Bank records
 - M-Pesa
 - Staff
 - Salaries
 - Approval workflows
 - Financial reports
 - Audit trails
-

51. Phase 4 — Projects, Grants & Partnerships

3-4 weeks

- Partner database
 - NGO management
 - Grant opportunities
 - Proposal pipeline
 - Project management
 - Project budgets
 - Project documents
 - Reporting
 - Partner communication
-

52. Phase 5 — Enterprise Expansion

Modules activated as MAKU's businesses mature:

- Honey
- Hive manufacturing
- Hides & skins
- Poultry
- Bones & horns
- Dairy
- Seedlings
- Value addition

This prevents MAKU from paying for complicated modules before they are needed.

53. Estimated Overall Timeline

A realistic initial implementation would be approximately:

16-24 weeks

for the core platform and major operational modules.

However, the system will be delivered in usable stages.

MAKU does not need to wait six months before seeing value.

For example:

Month 1: Member/CIG system + website

Month 2: Livestock operations

Month 3: Finance/admin

Month 4: Documents/projects/partners

Following months: Enterprise modules and integrations

Final timelines will depend on the agreed scope, availability of MAKU personnel, data preparation and feedback during development.

54. Data Migration

If MAKU currently has:

- Excel files
- Member lists
- CIG lists
- livestock records
- financial records
- documents

Trends CORE Solutions can assist in preparing and importing the information into the new system.

We recommend cleaning the data before migration.

Bad data entered digitally is still bad data.

55. Training

The system will be designed for ordinary users, not IT specialists.

Training will cover:

Management

- Dashboards
- Reports
- Approvals
- Governance

Staff

- Daily operations
- Records
- Payments
- Documents

Field Officers

- Mobile data entry
- Livestock
- Members
- CIGs

Administrator

- Users
- Roles
- Configuration
- Backups

56. Support & Maintenance

Following deployment, Trends CORE Solutions can provide:

- System maintenance
- Security updates
- Bug fixes
- Backups
- User support
- Training
- Feature enhancements
- Hosting management
- Integration support

A separate annual support and maintenance agreement can be established.

57. Recommended Development Model

Because MAKU has many ideas and activities, I recommend not locking the organization into a giant fixed system from day one.

Instead:

Build the foundation.

Deploy it.

Learn from actual MAKU operations.

Improve it.

Add modules as the cooperative grows.

This reduces risk and ensures the software reflects how MAKU actually works.

58. Indicative Cost Structure

I would recommend presenting the price in modules/phases, rather than one intimidating figure.

For example:

Component	Indicative Cost
Discovery & System Design	KES 80,000
Core Member & CIG System	KES 120,000
Livestock Management	KES 180,000
Finance & Administration	KES 180,000
Document Centre	KES 100,000
Projects & Grants	KES 100,000
Public Website	KES 100,000
Reporting & Dashboards	KES 80,000
Initial Deployment & Training	KES 60,000

Indicative Phase 1-4 investment: KES 1,000,000

This is something I would not finalize until you have done a proper discovery session with MAKU, because you are the sole developer and the scope can become very large.

We can instead produce a formal quotation after the discovery phase.

59. Hosting & Running Costs

Separate recurring costs should be clearly explained.

These may include:

- Cloud hosting
- Domain
- SSL/security
- Email
- SMS
- WhatsApp services
- M-Pesa transaction/integration costs
- Document storage
- Backups

Where third-party services charge MAKU directly, those charges should be distinguished from Trends CORE Solutions development fees.

60. Important Recommendation: Don't Build Accounting From Scratch

One piece of advice I would strongly recommend:

Do not attempt to recreate a full accounting package unnecessarily.

MAKU needs financial control, but a complete accounting system can become a major project by itself.

The first version should focus on:

Cash → Purchases → Sales → Expenses → Payments → Approvals → Reports → Audit trail

Then integrate with a proper accounting solution later if MAKU requires full accounting standards, payroll, tax processing and financial statements.

This will keep the project manageable for a sole developer.

61. Another Important Recommendation: Build Around Transactions

The heart of MAKU should be the transaction.

Almost everything should eventually connect to:

Who?

What?

When?

How much?

Why?

Who approved it?

What document supports it?

For example:

Farmer → Animal → Sale → Buyer → KES 45,000 → KES 200 service charge → Expenses → Farmer settlement → M-Pesa → Receipt

That creates a reliable digital trail.

62. MAKU Digital Identity

The website should also establish a stronger identity for MAKU.

Not just:

"MAKU is a livestock cooperative."

But:

MAKU is a farmer-led aggregation, livestock marketing and emerging agricultural value-addition organization based in Merti, Isiolo County.

Its activities can tell a much bigger story:

Livestock + Honey + Dairy + Poultry + Environment + Value Addition + Farmer Aggregation

That story is particularly important when approaching development partners.

63. The "MAKU Impact Map"

I would actually make this a signature feature.

The public website could eventually display a map of MAKU's operational footprint.

For example:

ISILOLO COUNTY

- CIG 01
- \
-
- CIG 04 --- MAKU HQ --- ● Feedlot
- |
- Collection
- |
- Borehole
- |
- Apiary
- |
- CIG 12

Clicking a location could show approved public information.

This gives NGOs and partners a visual understanding of MAKU's reach.

64. "MAKU Needs Support" Portal

Another outside-the-box idea:

Create a public section:

SUPPORT MAKU'S FARMERS

Projects requiring support could be displayed.

Example

Feedlot Borehole

Status: 70%

Required: KES XXX

Purpose:

Reliable water supply for livestock aggregation and feedlot operations.

Another:

Honey Production Expansion

Required:

- Hives
- Protective equipment
- Honey harvesting equipment
- Processing equipment

Another:

Tree Seedling Programme

Required:

- Seedling nursery equipment
- Water
- Seedlings
- Transport

Each project could have:

Problem → MAKU's Solution → What is Needed → Expected Impact → Contact

This gives partners a very clear entry point.

65. MAKU as a Data-Driven Cooperative

Over time, MAKU will accumulate something extremely valuable:

DATA.

Not personal data for its own sake, but operational information about:

- Farmers
- Livestock
- Markets
- Prices
- CIGs
- Production
- Environment
- Projects
- Sales
- Demand

With proper privacy controls, MAKU can eventually use this information to make better decisions.

For example:

Which CIG supplies the most livestock?

Which months produce the highest sales?

Which animals have the highest margins?

How much water does the feedlot consume?

Which projects have generated the greatest impact?

How many farmers benefit from each enterprise?

That turns the software into a management tool, not merely a digital filing cabinet.

66. Long-Term Vision

The long-term vision is:

One MAKU. One Digital Record.

A member should eventually be visible across the entire MAKU ecosystem.

For example:

Ahmed

Member

↓ CIG 07

↓ 12 livestock supplied

↓ 4 animals sold

↓ KES XXX received

↓ 2 beehives purchased

↓ 200 seedlings planted

↓ Milk supplied

↓ Membership contribution

↓ Share contribution

Everything belongs to the same member profile.

67. Why Trends CORE Solutions

Trends CORE Solutions develops practical digital solutions for businesses and organizations.

Our approach is based on:

Simplicity

Technology should make work easier, not more complicated.

Practicality

We build around the client's actual daily operations.

Scalability

The system grows as the organization grows.

Local Understanding

The platform is designed with the realities of Kenyan businesses and organizations in mind.

Long-Term Partnership

We don't simply deliver software and disappear.

We aim to become MAKU's technology partner as its operations develop.

68. Proposed Project Governance

MAKU should nominate a small implementation team.

MAKU

Project Sponsor: Chairman

Business Lead: Secretary General

Finance Lead: Treasurer

Operations Lead: Livestock/Operations Officer

System Champion: One designated staff member

Trends CORE Solutions

Lead Developer / Solution Architect

The MAKU team will validate workflows and provide feedback while Trends CORE Solutions handles system design and development.

69. Success Criteria

The project will be considered successful when MAKU can:

- ✓ Find any member quickly.
- ✓ Know which CIG a member belongs to.
- ✓ Record livestock without relying on notebooks.
- ✓ Track animals from farmer to sale.
- ✓ Calculate the KES 200 service charge automatically.
- ✓ Know how much each farmer should receive.
- ✓ Track purchases and sales.
- ✓ Track petty cash and expenses.
- ✓ Track approvals.
- ✓ Retrieve important documents quickly.
- ✓ Generate weekly and monthly reports.
- ✓ Manage projects and NGO opportunities.
- ✓ Communicate with members and partners.
- ✓ Present MAKU professionally online.
- ✓ Allow potential partners to easily find MAKU.

70. Final Outcome

At the end of the initial implementation, MAKU will have:

A digital member register

CIG management

Livestock marketing system

Feedlot management

Purchases & sales

Finance & expenses

Petty cash

Procurement

Document Centre

Project management

NGO/Partner management

Communication tools

Reports & dashboards

Public website

MAKU location/GIS

Secure user access

Audit trails

Foundation for future enterprises

71. Conclusion

MAKU is at an important stage of growth.

Its activities are expanding from livestock marketing into beekeeping, environmental conservation, feedlot management, hides and skins, poultry, dairy and other value-addition opportunities.

This growth creates both an opportunity and a challenge.

Without proper systems, growth can lead to:

- Lost records
- Poor accountability
- Slow reporting
- Difficult audits
- Disorganized documents
- Missed partnership opportunities
- Weak financial visibility

The proposed MAKU Digital Cooperative Platform is intended to address these challenges by giving MAKU one centralized system for managing its members, operations, finances, documents, projects and partnerships.

More importantly, it will provide the digital foundation for MAKU's next stage of growth.

TRends CORE Solutions

We build solutions for business.

Proposed Solution: MAKU Digital Cooperative Platform Client: Merti Animal Key Users (MAKU)

Location: Merti, Isiolo County, Kenya

Technology should not replace the people who run MAKU. It should give them better tools to run it.

One thing I would change before you send this to MAKU

I would not send the KES 1M figure yet as a final quotation. The scope you've described is actually large enough to become a multi-year platform. Since you're the sole developer, the biggest risk is you accidentally selling yourself a 12-18 month headache.

I recommend we turn this into a professional 12-15 page client proposal + separate commercial quotation, with:

1. Cover page
2. MAKU's current situation
3. Proposed digital vision
4. System architecture visual
5. Modules and workflows
6. Website concept
7. NGO/partner strategy
8. Technology stack
9. Implementation phases
10. Timeline
11. Training/support
12. Investment & payment schedule

13. Assumptions/exclusions

14. Ownership/IP

15. Acceptance/sign-off

And importantly, we should define Phase 1 very tightly so you, as the sole developer, don't get trapped by "while you're at it, can you also add..." scope creep.



MAKU Digital Cooperative Platform

Prepared by Trends CORE Solutions

Technology should not replace the people who run MAKU.

It should give them better tools to run it.